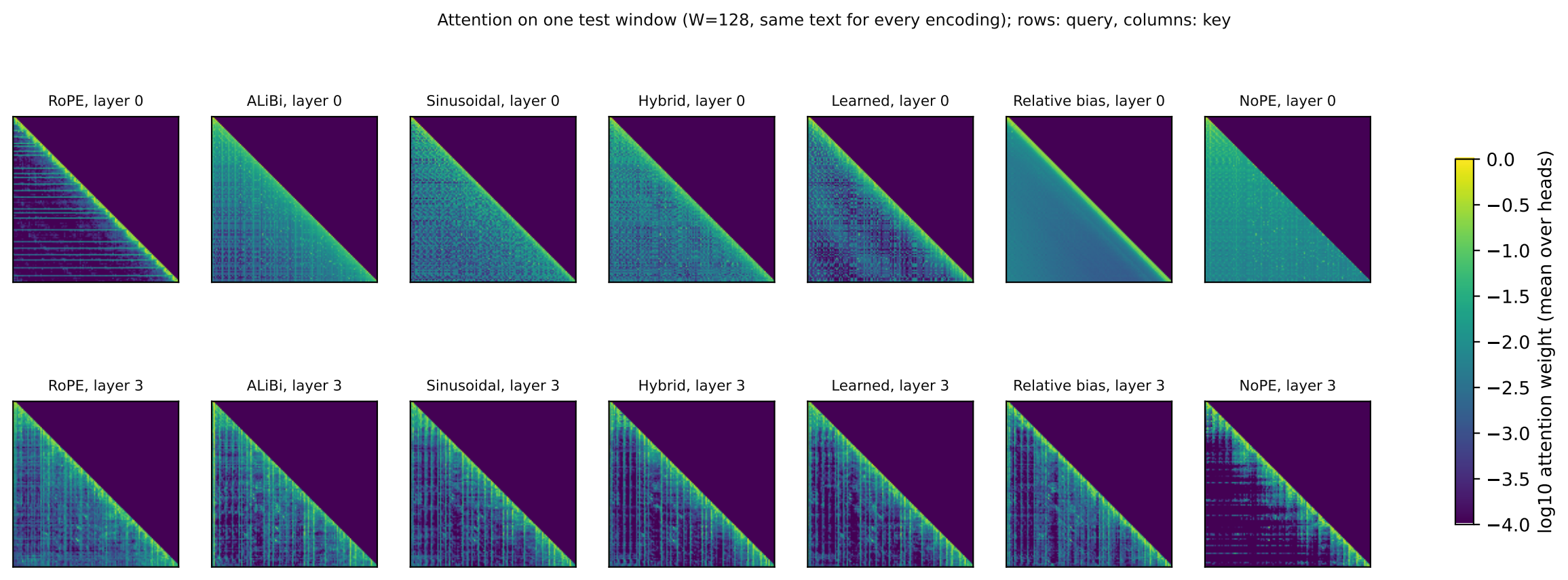

Attention on one test window (W=128, same text for every encoding); rows: query, columns: key
RoPE, layer 0
ALiBi, layer 0
Sinusoidal, layer 0
Hybrid, layer 0
Learned, layer 0
Relative bias, layer 0
NoPE, layer 0
RoPE, layer 3
ALiBi, layer 3
Sinusoidal, layer 3
Hybrid, layer 3
Learned, layer 3
Relative bias, layer 3
NoPE, layer 3
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
log10 attention weight (mean over heads)